

CAN 304

Computer Systems Security

Tutorial 3. Identifying A Research Topic on Security

Week 3: 2026-03-19, 14:00-15:00, Thursday

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Top Conferences on Security

- **Security**

- S&P: IEEE Symposium on Security and Privacy (<https://sp2026.ieee-security.org/index.html>)
- USENIX Security Symposium (<https://www.usenix.org/conference/usenixsecurity26>)
- CCS: ACM Conference on Computer and Communications Security (<https://www.sigsac.org/ccs/CCS2026/index.html>)
- NDSS: Network and Distributed System Security Symposium (<https://www.ndss-symposium.org>)

- **Cryptography**

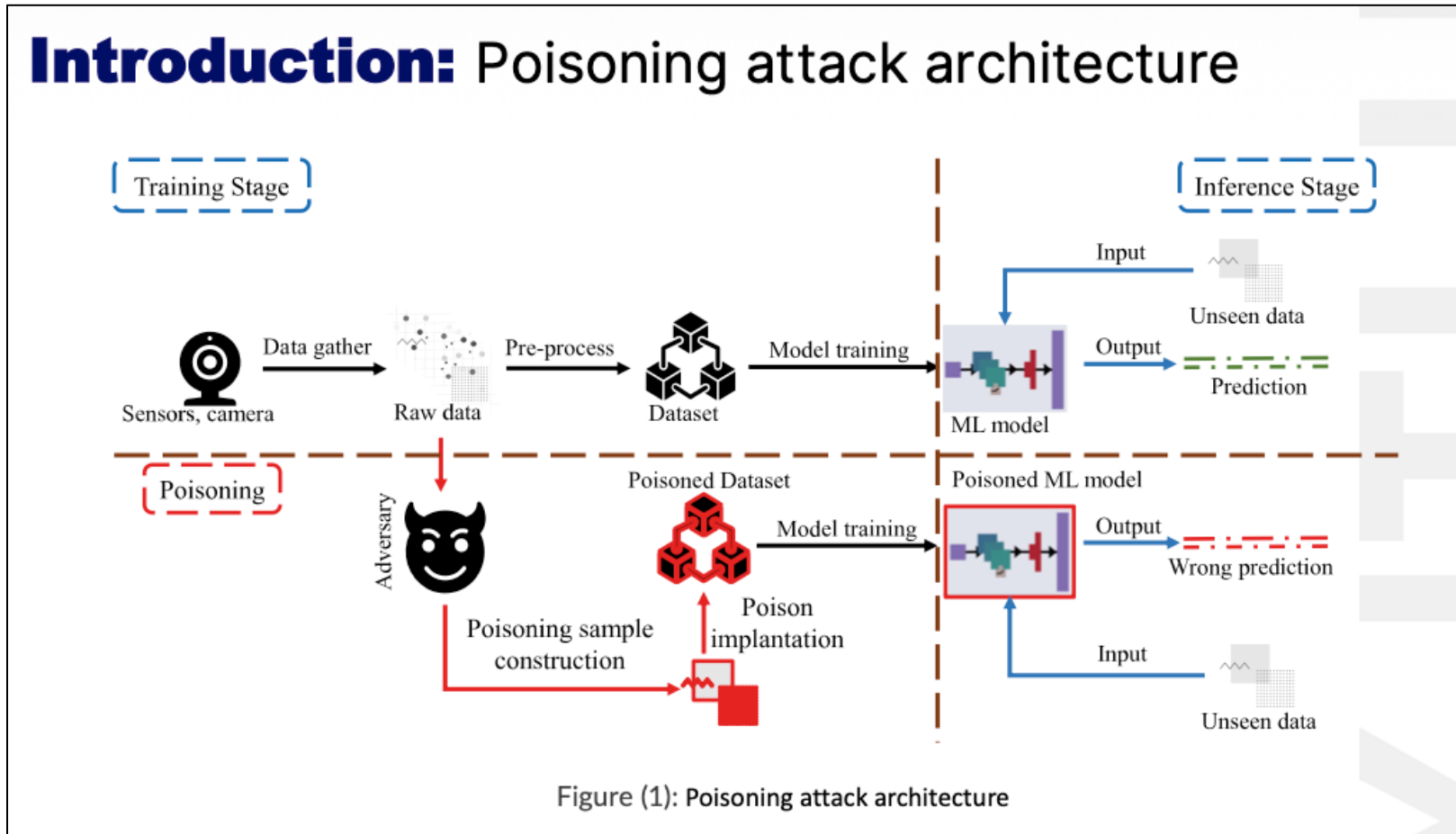
- Eurocrypt <https://eurocrypt.iacr.org/2026/>
- Crypto <https://crypto.iacr.org/2026/>
- Asiacrypt <https://asiacrypt.iacr.org/2025/>

Outstanding Project in Past Years

- Random, Latent and Salient Backdoor Attacks on Deep-Learning Models
 - 22/23/S2
 - Published in 2024 International Joint Conference on Neural Networks (IJCNN)
 - Lyu, Zihan, Dongheng Lin, Shengkai Sun, Jie Zhang, and Ruiming Zhang. "SPSS: A Saliency-based Poisoning Selection Strategy for Selecting Backdoor Attack Victims." In 2024 International Joint Conference on Neural Networks (IJCNN), pp. 1-9. IEEE, 2024.

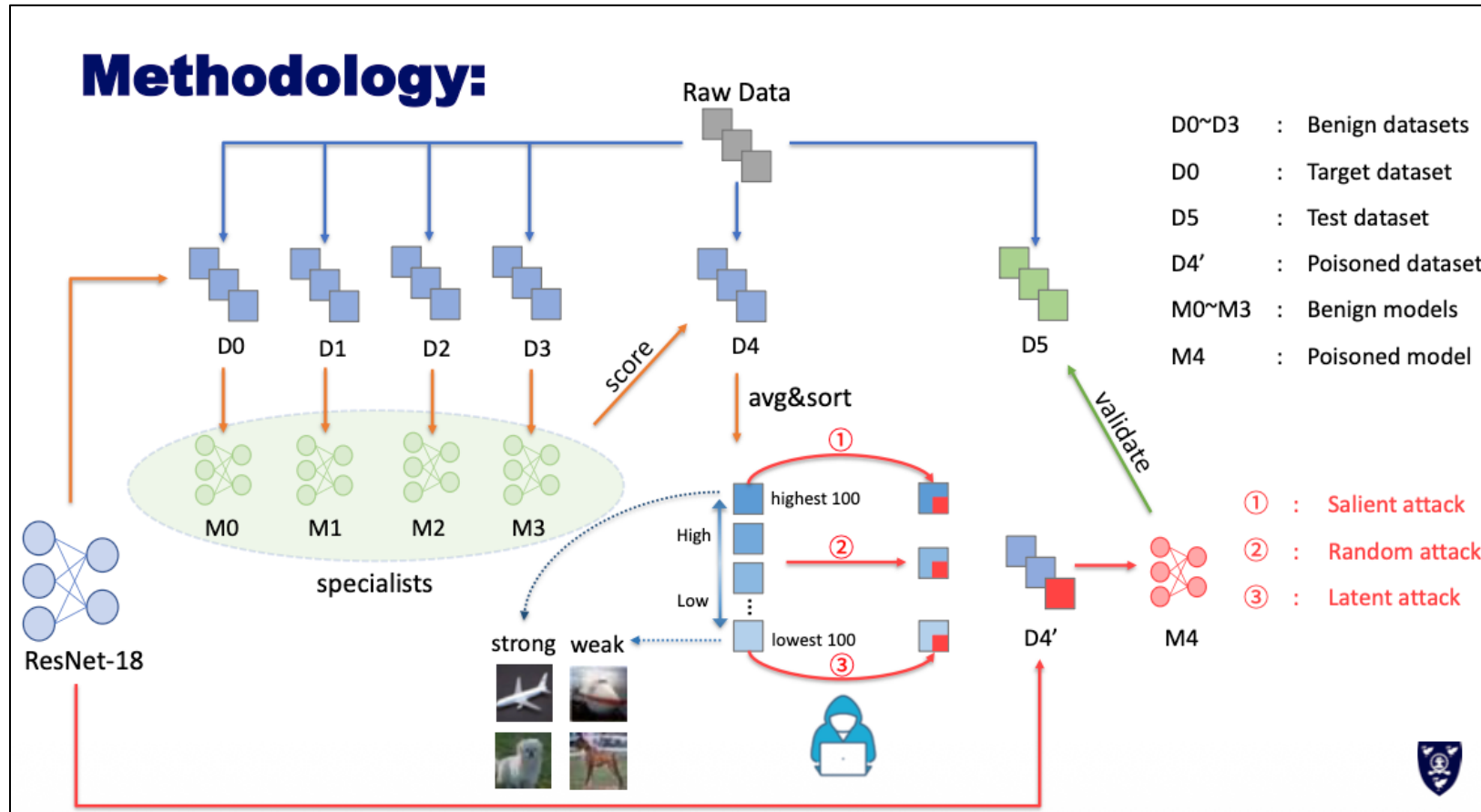
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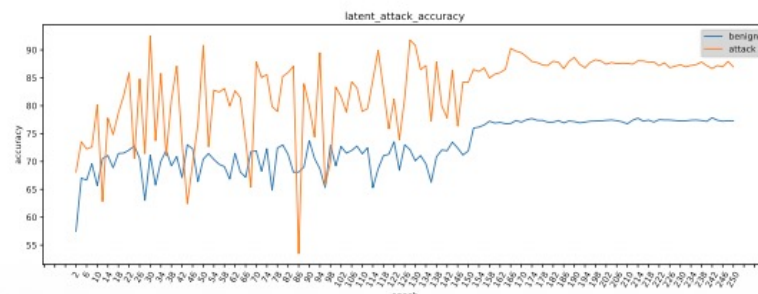
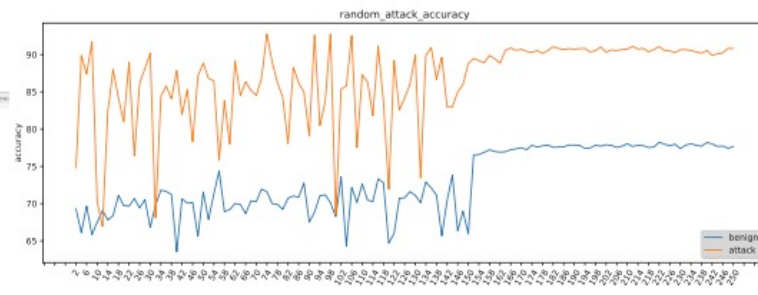
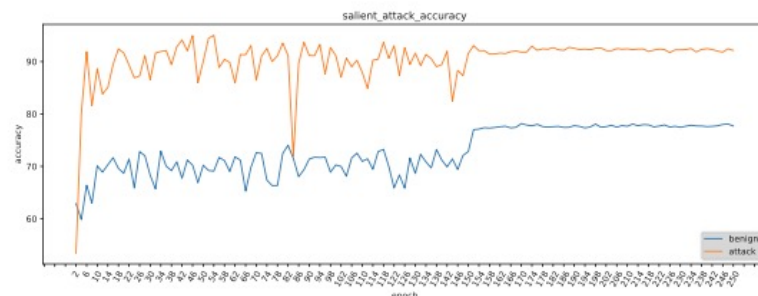
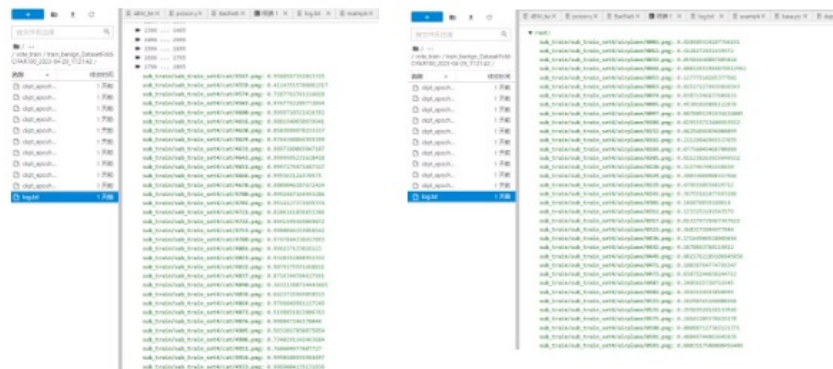


Outstanding Project in Past Years

- Random, Latent and Salient Backdoor Attacks on Deep-Learning Models
 - 22/23/S2

Experimental Results:

Attack Methods	Random	Latent	Salient
Accuracy Score	90%	88%	93%



Outstanding Project in Past Years

- TrueSeeing: Defensive software against text-encoding based invisible attacks
 - 23/24/S2

State-of-the-art attack methods

Invisible Characters You cannot see some characters that a program can see.

Homoglyphs Different characters that look the same (the introduction).

Reordering Texts whose order looks different to you and to a computer program.

Deleteion Something you can see cannot be seen by a computer program.

Outstanding Project in Past Years

- TrueSeeing: Defensive software against text-encoding based invisible attacks
 - 23/24/S2

Demonstration: TrueSeeing - Encoding Format

Choose Encoding format

Encoding Format

Text Examination

Signature Production

Select one encoding format:

☒ UTF-8

☐ UTF-16

Select

tips

You have to choose one encoding format from UTF-8 and UTF-16.

Revealing Malicious Characters

Encoding Format

Text Examination

Signature Production

Original Text

google.com
google.com
osLEJSxyp v a
The title is "مفتاح معايير الويب" in Arabic.
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Load from system clipboard

Load from local file

tips

Input the text from your system clipboard or from the local file.
The original appearance of the input shown in "Original Text".

True Text

google.comrngoogle.comrnosLEJSxyp v arinThe title is "مفتاح معايير الويب"u200c in Arabic.rin2
u20623u20641

Digital Signature

Encoding Format

Text Examination

Signature Production

Generated Signature

6f14ccca2e54c4684d1dcda6767dfdbde77cb6a1d42e9a1ea364e2b283863b541931df14
d796440cc67a4c7e90de7d4fe588b0db2a90d9922c5a73bd21cc6a49422faeaff0cee0485
eedecdf30be98644a2bf7011ecaff6fd13095f31a42fe9d7f6b44698eff77b528b170f00591
b5d20b14e3395459791b942bb57996131caf49b03de935a236231ac49f36d0a91a06a3c
e35013724924c7a49e7db7fab98b7ae7a0ae0fe740ff4d1462d308e5400ee4964e292b8
2827005298e3e2df04fc72a1e3dd337d0bea7f34dcac84319b8146fdce3c8f90573a88171
3f9e7bfec91fcb0713044ec3e3d6ddbbaee976bf273e86837408ce33cd0ec04a6

Successfully generate the signature for this text
in the above text area with the algorithm:
RSA-FDH: RSA full-domain hash.

tips

Find the file named "private.pem" for private key and "public.pem" for public key.



Invited Talk

- Yuxin Xia
 - PhD started in Sep 2024
 - Research interests: Public-Key Cryptography, specifically focus on Key Exchange, Digital Signature
 - Topic: Revocable Signature: Handling Valid but Unauthorized Non-Fungible Token through Auxiliary Embedded Key
- Ziyang Yang
 - PhD started in Jan 2025
 - Research interests: AI Security, specifically focus on Privacy of ML
 - Topic: Privacy-Preserving Machine Learning: A Co-design Approach for Training & Inference